# Vector Fields and Line Integrals

**ANSWER KEY**

## Section 1: Vector Fields, Divergence, and Curl

**1** Compute the divergence $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = \langle x^2 y,\ 3yz,\ -2xz \rangle$.

**a.** $2xy + 3z - 2x$ ✓

**b.** $2xy + 3y - 2z$

**c.** $x^2 + 3z - 2x$

**d.** $0$

$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(3yz) + \frac{\partial}{\partial z}(-2xz) = 2xy + 3z - 2x.$

**2** Compute the 2D curl of $\mathbf{F} = \langle -y,\ x \rangle$.

**a.** $2$ ✓

**b.** $0$

**c.** $1$

**d.** $-2$

$\text{curl}(\mathbf{F}) = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) = 1 - (-1) = 2.$

**3** If a vector field has $\nabla \times \mathbf{F} = \mathbf{0}$ throughout a simply connected domain, what is it called?

**a.** **Conservative (or irrotational)** ✓

**b.** Incompressible

**c.** Solenoidal

**d.** Harmonic

A curl-free vector field on a simply connected domain is conservative, meaning $\mathbf{F} = \nabla f$ for some scalar potential function $f$.

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## Section 2: Line Integrals with Respect to Arc Length

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**4** State the formula for $\int_C f(x,y)\,ds$ along a curve parameterized by $\mathbf{r}(t)$ for $a \le t \le b$.

**a.** $\int_a^b f(x(t),y(t))\sqrt{[x'(t)]^2+[y'(t)]^2}\,dt$ ✓

**b.** $\int_a^b f(x(t),y(t))\,dt$

**c.** $\int_a^b f(x(t),y(t))[x'(t)+y'(t)]\,dt$

**d.** $\int_a^b f'(t)\,dt$

Since $ds = \|\mathbf{r}'(t)\|dt = \sqrt{(x')^2+(y')^2}\,dt$, the scalar line integral is $\int_a^b f(\mathbf{r}(t))\|\mathbf{r}'(t)\|dt$.

**5** Evaluate $\int_C (x+y)\,ds$ where $C$ is the line segment from $(0,0)$ to $(3,4)$.

**a.** $\frac{35}{2}$ ✓

**b.** $35$

**c.** $7$

**d.** $12$

Parametrize: $x = 3t$, $y = 4t$ for $t \in [0,1]$. $ds = \sqrt{9+16}\,dt = 5\,dt$.
$\int_0^1 (3t+4t)(5)\,dt = 35\int_0^1 t\,dt = 35[t^2/2]_0^1 = 35/2$.

**6** Find the mass of a wire bent in the shape of the semicircle $\mathbf{r}(t) = \langle 2\cos t, 2\sin t\rangle$ $(0 \le t \le \pi)$ with density $\rho(x,y) = y$.

**a.** $4$ ✓

**b.** $8$

**c.** $2\pi$

**d.** $4\pi$

$ds = 2\,dt$. $M = \int_0^{\pi} (2\sin t)(2)\,dt = 4[-\cos t]_0^{\pi} = 4[-(-1)-(-1)] = 4(2) = 8$.

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## Section 3: Line Integrals of Vector Fields (Work Integrals)

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**7** Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, -x \rangle$ along the circle $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$ for $0 \le t \le 2\pi$.

**a.** $-2\pi$ ✓

**b.** $2\pi$

**c.** $0$

**d.** $\pi$

$\mathbf{F}(\mathbf{r}(t)) = \langle \sin t, -\cos t \rangle$. $d\mathbf{r} = \langle -\sin t, \cos t \rangle dt$. Dot product: $-\sin^2 t - \cos^2 t = -1$. $\int_0^{2\pi} (-1)\, dt = -2\pi$.

**8** Find the work done by the constant force $\mathbf{F} = \langle 3, 5 \rangle$ moving an object along the straight line from $(1, 1)$ to $(4, 5)$.

**a.** $29$ ✓

**b.** $24$

**c.** $32$

**d.** $15$

Displacement $\mathbf{d} = \langle 4-1, 5-1 \rangle = \langle 3, 4 \rangle$. Work $= \mathbf{F} \cdot \mathbf{d} = (3)(3) + (5)(4) = 9 + 20 = 29$.

**9** Evaluate $\int_C (x\, dx + y\, dy)$ along the parabolic arc $y = x^2$ from $(0, 0)$ to $(1, 1)$.

**a.** $1$ ✓

**b.** $1/2$

**c.** $2/3$

**d.** $0$

Notice $\mathbf{F} = \langle x, y \rangle = \nabla(\frac{1}{2}x^2 + \frac{1}{2}y^2)$. By FTLI: $f(1,1) - f(0,0) = (1/2 + 1/2) - 0 = 1$.

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## Section 4: Fundamental Theorem of Line Integrals and Green's Theorem

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**10** State the Fundamental Theorem of Line Integrals for a conservative field $\mathbf{F} = \nabla f$.

**a.** $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ ✓

**b.** $\int_C \nabla f \cdot d\mathbf{r} = 0$ for all curves

**c.** $\int_C \nabla f \cdot d\mathbf{r} = \|\nabla f\|$

**d.** $\int_C \nabla f \cdot d\mathbf{r} = f'(\mathbf{r}(b))$

The work done by a conservative vector field depends only on the endpoints: $\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$.

**11** State Green's Theorem relating a line integral along a positively oriented boundary $\partial D$ to a double integral over $D$.

**a.** $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$ ✓

**b.** $\oint_C (P\,dx + Q\,dy) = \iint_D (P + Q)\,dA$

**c.** $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$

**d.** $\oint_C (P\,dx + Q\,dy) = 0$

Green's Theorem connects circulation around a closed curve to the flux of the curl across the enclosed region: $\oint_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA$.

**12** Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where $C$ is the unit circle $x^2 + y^2 = 1$ oriented counterclockwise.

**a.** $2\pi$ ✓

**b.** $8\pi$

**c.** $0$

**d.** $\pi$

$P = 3y, Q = 5x \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 5 - 3 = 2$. By Green's Theorem: $\iint_D 2\,dA = 2 \times \text{Area}(D) = 2(\pi \cdot 1^2) = 2\pi$.

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